

🔥 DAY 20 — STREAMING OUTPUT (REAL-TIME ANSWERS)

✅ GOAL

Upgrade from:

Wait → Full answer ❌

to: Token-by-token streaming ✅

(like ChatGPT typing)

✅ IMPORTANT

👉 You are using: `ollama.chat()`

✅ It **supports streaming** 👉 We will enable it

✅ ✅ **STEP 1 — UPDATE LLM:** 📄 `models/llm.py`

🔧 **FIND THIS FUNCTION:** `def generate(prompt):`

✅ REPLACE FULL FUNCTION

```
import ollama
```

```
def generate_stream(prompt):
```

```
    stream = ollama.chat(
        model="qwen:latest",
        messages=[{"role": "user", "content": prompt}],
        stream=True
    )
```

```
    for chunk in stream:
```

```
        content = chunk["message"]["content"]
```

```
        yield content
```

✅ KEEP OLD FUNCTION ALSO (DO NOT DELETE)

```
def generate(prompt):
```

```
response = ollama.chat(
    model="qwen:latest",
    messages=[{"role": "user", "content": prompt}] )
return response["message"]["content"]
```

✅✅ **STEP 2 — UPDATE API:** 📄 app/api.py

🔧 **ADD IMPORT**

```
from fastapi.responses import StreamingResponse
from models.llm import generate_stream
```

🔧 **FIND ENDPOINT**

```
@app.post("/ask")
async def ask(q: Query):
```

✅ **REPLACE FULL FUNCTION**

```
@app.post("/ask")
async def ask(q: Query):
    async def stream():
        # ✅ run agent (still normal)
        answer = await agent.run(q.query)
        # ✅ stream final answer
        for token in generate_stream(answer):
            yield token
    return StreamingResponse(stream(), media_type="text/plain")
```

✅✅ **STEP 3 — IMPORTANT NOTE**

👉 Your agent still: generate(final_prompt)

✅ That is OK 👉 Streaming happens at API level

✓ ✓ FINAL FLOW NOW

User Request

↓

Agent computes answer

↓

Streaming starts ✓

↓

User sees answer typing

✓ ✓ TEST

Run: `uvicorn app.api:app --reload`

✓ Test in browser:

`http://127.0.0.1:8000/docs`

Call /ask

✓ Expected output:

Answer appears gradually ✓

✓ ✓ WHAT YOU BUILT TODAY

Real-time AI response system

✓ ✓ SYSTEM LEVEL NOW

✓ AI system

✓ API + Docker

✓ Async execution

✓ Caching

✓ Memory (vector)

✓ ✓ Streaming output

✅✅ INDUSTRY TERM

Token Streaming / Real-time inference

Used in:

- ChatGPT
 - Copilot
 - Claude
-

✅✅ IMPORTANT (TRUTH)

👉 This is exactly: How ChatGPT shows typing effect ✅

✅✅ SMALL LIMITATION

Currently: Agent completes → THEN streaming starts

👉 Next level (later): True streaming from LLM inside agent

✅✅ NEXT

Day 21 → Monitoring + Logging (latency, debug, tracing)

If ready: 👉 say **Day 21**